University of Cape Town
Department of Computer Science

Computer Science CSC3003S
Final Examination, November 2015

Read through the entire examination paper before you start writing so that you can answer the easy questions first. The relative importance of a question is indicated by the marks allocated to it. Please write legibly.

This paper consists of Four Sections:

Section 1: Software Design — compulsory
Section 2: Compilers — compulsory
Section 3: Theory of Algorithms — compulsory
Section 4: Choice Section — Choice of 3 out of 6 questions.

You must answer all questions for each of Section 1 to Section 3 (that is answer all questions from Question 1 to Question 6), and answer three questions from the six in the Choice Section (Section 4).

Total Marks: 90	Approximate marks per question are shown in brackets.
Time: 3 hours + 10 minutes reading	The use of calculators is permitted.

Please write your answers in three separate books. Use one book for each of Software Design, Compilers and Theory of Algorithms. When you answer questions from the choice section please put them with the corresponding answers from the compulsory sections.

So, for example, all Compilers questions go into answer book 2 as well as any choice questions about compilers that you choose to answer.

Your attention is drawn to the instructions on your answer book on the front and the back. In particular "**EVERY CANDIDATE MUST enter the book number and the number of each question answered (in the order in which it has been answered)**" on the table provided on the front cover of the first book.

Please **do this for the front cover of the different books used for each section.**

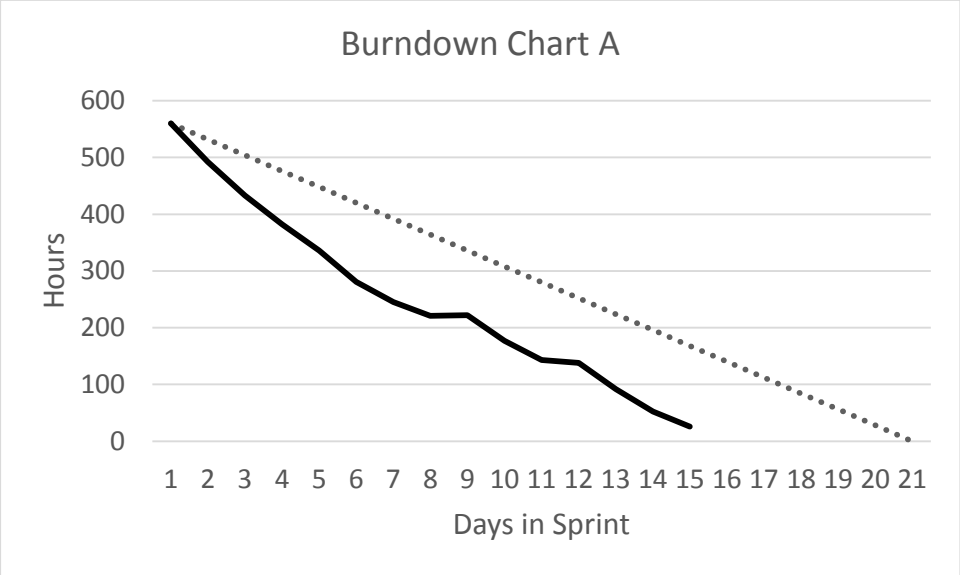
This is needed so that we can allocate marks and not miss questions. A list of sub-questions is *not* required.

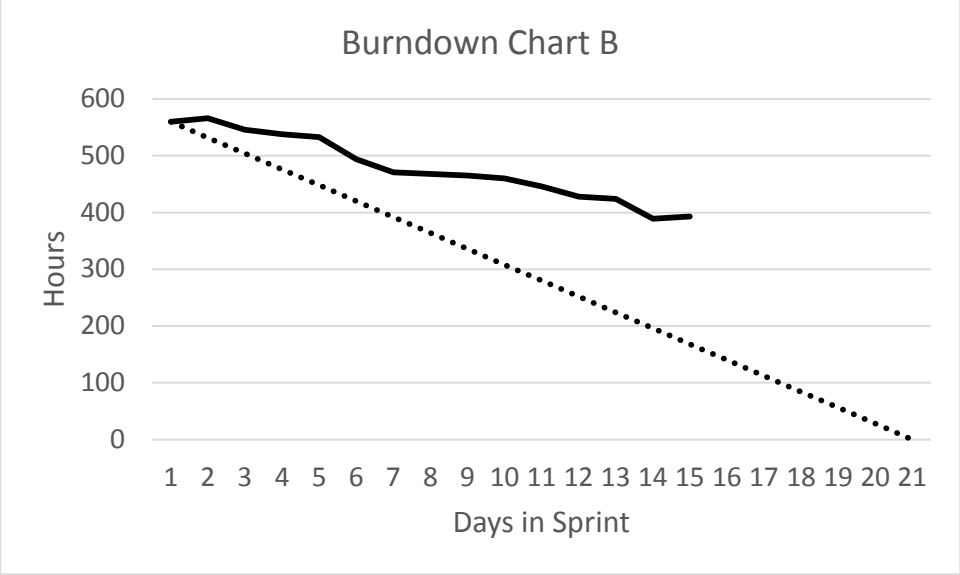
Section 1. Advanced Software Design [22 Marks]

Question 1. **[9]**

a. In a scrum project management process, when does the development team reflect on how well they have been working and decide on potential improvements to how they go about things? How is this reflection process run? [1]

Consider the following two burndown charts ('A' and 'B') taken from a twenty-one day scrum sprint that starts off with an estimated 560 hours of work. The dotted lines simply indicate a line that counts down from that towards zero on day twenty-one. The team has just completed day fifteen and they been working solidly on their sprint backlog and there have not been any external bottlenecks. Consider yourself to be in the role of scrum master.





If you think you have not been provided with enough information to answer the questions below then indicate what is missing in the answer and say what assumptions you need to make in order to provide an answer.

b. What does the solid line indicate? [1]

c. What is happening in the sprint illustrated by 'A'? As scrum master give an outline and justification for any action that you may consider at this point? [2]

d. What is happening in the sprint illustrated by 'B'? As scrum master give an outline and justification for any action that you may consider at this point? [2]

e. If you (the scrum master) decide there are problems in the way you and your team have been working *when* and *how* might you address them? What sort of things will need to be addressed with respect to sprint 'A'? And with sprint 'B'? [3]

Question 2. **[13]**

a. What is meant by software architecture? [2]

b. Many architectural patterns set out to achieve "high cohesion" and "low coupling". Why is this so desirable? [2]

c. What is the difference between using the GPL (GNU General Public License) for software and putting the code in the public domain? [3]

d. Much of the course has been about software development project management. When it comes to open source software development is the management required very different? Discuss the need, appropriate style and key objectives of open source management. [6]

Section 2. Compilers [22 Marks]

Question 3. **[16]**

a. Various data structures are used in the process of compiling and executing programming language code. Indicate when each of the following data structures are used and what they are used for:

i. Abstract Syntax Tree [3]

ii. Control Flow Graph [2]

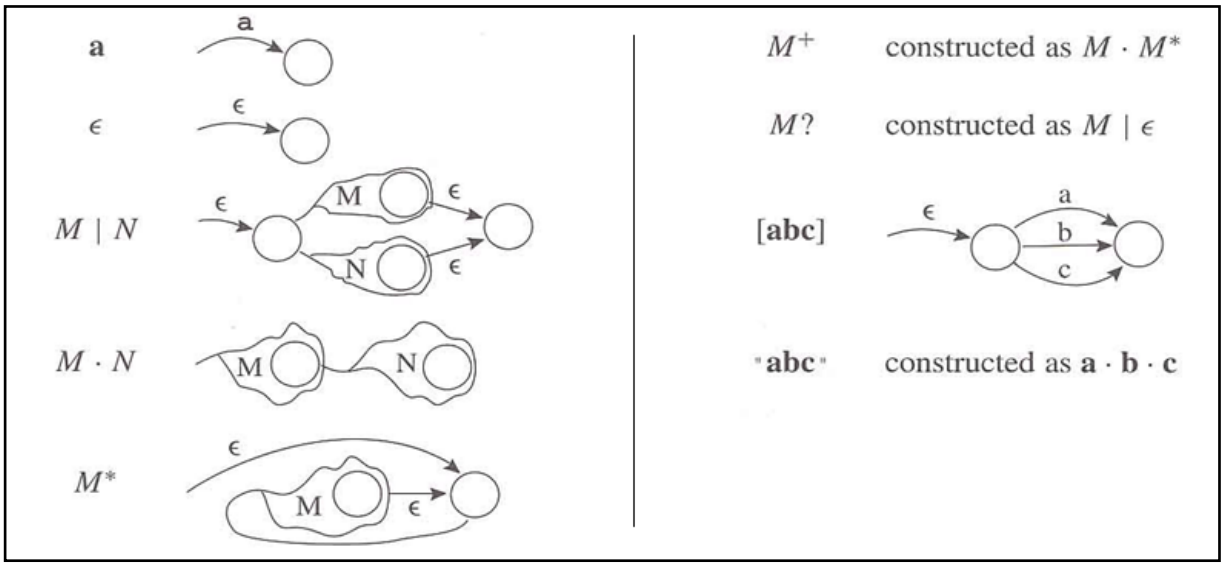
iii. Activation Record [2]

b. Consider the following Java program which contains 2 unique errors. Indicate which line the error is on as identified on the left, what the error is and also indicate which phase of the compilation process the error is identified - Lexical Analysis, Parsing(Syntactic Analysis) or Semantic Analysis. [4]

```
1. import java.util.*;
2.
3. class SumSquares {
4.
5.    static int calculateSumSquares(int n) {
6.       int sumSquares = 0;
7.       for (int i=1; i<=n; i++) {
8.          sumSqaures += i*i;
9.       }
10.      return sumSquares;
11.   }
12.
13.   public void static main(String args[]) {
14.      Scanner kb = new Scanner(System.in);
```

```
15.       System.out.print("Enter number:");
16.       int num = kb.nextInt();
17.       int sumSquares = calculateSumSquares(num);
18.       System.out.println("Sum squares is " + sumSquares);
19.    }
20.
21.}
```

c. Convert the following Regular Expression to a Finite Automata, using the rules defined below: $a^{+}(a|b)$ [5]



Question 4. **[6]**

Consider the following program for Code Block Reorganisation:

```
Start:  Statement0
        Jump C
A:      Statement1
B:      Statement2
        Jump D
C:      Statement3
        Jump A
D:      Statement4
        Done:
```

a. Separate the code into basic blocks [1]

b. Rearrange the blocks into a trace. [1]

c. Optimise the code by removing redundant jumps. [1]

d. Rewrite the following code fragment below to a more optimal form using the optimisation techniques inlining and loop unrolling code. [2]

```
int cube(int n) {
     return(n*n*n);
}
int cube_sum = 0;
```

```
for(int i=2; i<=4; i++) {
      cube_sum += cube(i);
}
print(cube_sum);
```

e. At the end of the Instruction Selection process the generated code has "holes". Explain why these holes exist. [1]

Section 3. Theory of Algorithms [22 Marks]

Question 5. **[9]**

a. What does the following algorithm do? [1]

```
ALGORITHM mystery(int x, int y)
        while y ≠ 0 do
          t ← x mod y
          x ← y
          y ← t
        return x
```

b. Explain your answers in each case below:

i. Is $O(n!) \subseteq O(2^n)$? [1]

ii. Is $O(n \log_2 n) \subseteq O(n!)$? [1]

iii. Is $\Theta(n^3) \subseteq \Theta(n^2)$? [1]

c. Consider the following algorithm:

```
ALGORITHM whoknows(int x, int y, int z, int a)
  if a > 0 do
      whoknows(x,z,y,a-1)
      print x, z, a
      whoknows(y,x,z,a-1)
```

i. What does the algorithm above do? [1]

ii. Set up a recurrence relation for the number of basic operations being performed and solve it. Show all your work. [3]

d. Consider the two algorithms below. Are they in the same efficiency class or not? Justify your answer [1]

```
ALGORITHM A2(int n > 0)
if n=1 return 1 else return 1 + A2(n/2)

ALGORITHM A5(int n > 0)
if n=1 return 1 else return 1 + A5(n/5)
```

Question 6. **[13]**

a. Strassen's algorithm for matrix multiplication makes use of the fact that two 2x2 matrices can be multiplied as below. Use this result to write an algorithm in pseudocode for matrix multiplication of an n x m matrix by an m x k matrix (yielding an n x k matrix). [4]

$$\begin{bmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{bmatrix} \times \begin{bmatrix} b_{00} & b_{01} \\ b_{10} & b_{11} \end{bmatrix} = \begin{bmatrix} c_{00} & c_{01} \\ c_{10} & c_{11} \end{bmatrix}$$

$$\begin{bmatrix} m_1+m_4-m_5+m_7 & m_3+m_5 \\ m_2+m_4 & m_1+m_3-m_2+m_6 \end{bmatrix}$$

$$\begin{aligned} m_1 &= (a_{00}+a_{11})*(b_{00}+b_{11}) \\ m_2 &= (a_{10}+a_{11})*b_{00} \\ m_3 &= a_{00}*(b_{01}-b_{11}) \\ m_4 &= a_{11}*(b_{10}-b_{00}) \\ m_5 &= (a_{00}+a_{01})*b_{11} \\ m_6 &= (a_{10}-a_{00})*(b_{00}+b_{01}) \\ m_7 &= (a_{01}-a_{11})*(b_{10}+b_{11}) \end{aligned}$$

.

b. Which is better suited for finding the shortest path between two edges in a graph, depth-first search or breadth-first search? Explain your answer using a representative example. [2]

c. Consider the algorithm below. What is the efficiency class of this algorithm in the worst case? Explain what the input will look like in the worst case [2]

```
ALGORITHM X(A[0..n-1])
  for i=1 to n-1
    v = A[i]
    j = i-1
    while j ≥ 0 and A[j]>v do
      A[j+1] = A[j]
      j = j - 1
    A[j+1] = v
```

d. Prove that the Hamiltonian Circuit Problem (HCP) is in the class NP. Suppose you come up with an algorithm tomorrow that solves HCP and runs in polynomial time. Explain the consequences of such a discovery. [3]

e. Given a text and a pattern P[0..m-1], complete writing these two rules for creating the shift table, T, for the Horspool string searching algorithm. [2]

```
T(c) = … for a letter, c, in P [0..m-2]
T(c) = … for a letter not in P[0..m-2]
```

Section 4. Choice Section: Answer any THREE of the following six questions (0–Question 12). [24]

If you answer Question 7 or Question 8 (or both) then please put your answers in the same book with those of Section 1.

Question 7. ASD – Modern Vehicle Network [8]

The wiring in a modern car and truck is based on a digital network architecture. This replaces the electrical system with its relays and fuses. Such networks are simpler to install and weigh less. The key requirement is to achieve great reliability, hence a great deal of investigation has already been done into exactly what is needed.

The system has a number of intelligent subsystems (microcontrollers) such as electronic stability control and anti-lock braking which are very important safety systems for the automobiles. Other systems include engine management systems to reduce emissions and provide reliable performance.

These subsystem will probably have to be replaced as technology develops further. It is likely that the subsystems will be sourced from different subcontractors.

a. Present your architecture for such a system and justify your choices. In particular explain how this architecture supports reliability and allows for the independent development of some of the subsystems. [5]

b. What software development method would you recommend to develop this system? Why? [3]

Question 8. ASD – Waterfall [8]

What are the key advantages of the waterfall model? Consider the projects below and decide for each one of them if it would be the most suitable method. Explain why you say so.

i. The software system for guiding the next spacecraft to be sent to Pluto.

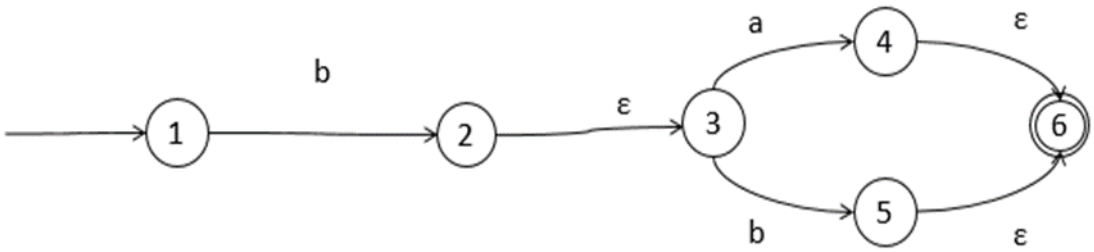
ii. A record-keeping system for a small advertising business.

iii. An optimising compiler for Haskell, providing many language extensions.

If you answer Question 9 or Question 10 (or both) then please put your answers in the same book with those of Section 2.

Question 9. Compilers – Front End [8]

a. Convert the following Nondeterministic Finite Automata (NFA) to a Deterministic Finite Automata. For each state in the DFA indicate from which NFA states it was derived. [4]



Consider the following grammar and answer the following questions regarding First and Follow Sets.

```
E  → T E'
E' → + T E' | ε
T  → F T'
T' → * F T' | ε
F  → ( E ) | id
```

b. What is the First(E)? [1]

c. What other non-terminal, other than T, has the same First Set as E? [1]

d. What is the First(E')? [1]

e. What is the Follow(E)? [1]

Question 10. Compilers – Back end [8]

a. Indicate whether the following statement is True or False. "During the Canonicalisation process ESEQ statements are moved towards the bottom of the IR tree, i.e. towards the leaves." [1]

b. Consider the following list of Canonicalisation rules:

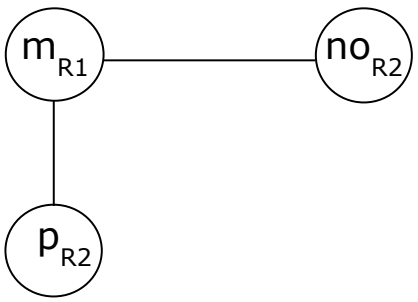
```
ESEQ(s1, ESEQ(s2, e))  =>  ESEQ(SEQ(s1,s2), e)
MEM(ESEQ(s, e1))  =>  ESEQ(s, MEM(e1))
JUMP(ESEQ(s, e1))  =>  SEQ(s, JUMP(e1))
CJUMP(op, ESEQ(s, e1), e2, l1, l2)  =>  SEQ(s,
CJUMP(op,e1,e2,l1,l2))
MOVE(ESEQ(s, e1), e2)  =>  SEQ(s, MOVE(e1, e2))
CJUMP(op, e1, ESEQ(s, e2), l1, l2)  =>
    SEQ(MOVE(TEMP t, e1), SEQ(s, CJUMP(op,TEMP t, e2, l1, l2)))
```

Transform this IR tree as much as possible, using the above rules. Show each rule being applied as a separate step. [3]

```
CJUMP(op, MEM(ESEQ(s1, ESEQ(s2, e))), e2, l1, l2)
```

c. Consider the following piece of code and the resulting graph colouring assigning registers during the process of Register Allocation. Rewrite the code by applying the appropriate register allocation, specify what other simple optimisation can be performed and apply it to the code. [4]

```
m=4
n=2
o=m/n
n=o
p=m+o
p=m*p
```



If you answer Question 11 or Question 12 (or both) then please put your answers in the same book with those of Section 3.

Question 11. ToA – Efficient closest points [8]

Consider the following pseudo-code to find the shortest distance between a set of points on a Euclidean plane, each with an x and y coordinate.

```
1.  Sort the points by x order (tiebreak on the y value) into P.
2.  Sort the points by y order (tiebreak on the x value) into Q.
3.  EfficientClosestPair(P, Q)
4.  EfficientClosestPair(P, Q):
5.    if n <= 3: return minimal distance found by brute force
6.    copy the first ceiling[n/2] points of P to Pl
7.    copy the same points of Q to Ql
8.    copy the remaining floor[n/2] points of P to Pr
9.    copy the same points of Q to Qr
10.   dl = EfficientClosestPair(Pl, Ql)
11.   dr = EfficientClosestPair(Pr, Qr)
12.   d = …
13.   m = P[ceiling(n/2)]-1].x
14.   copy all points of Q for which |x-m|<d into S[0..num-1]
15.   dminsq = d * d
16.   for i = 0 to num – 2:
17.     k = i + 1
18.     while k <= num – 1 and (S[k].y – S[i].y)^2 < dminsq:
19.       dminsq = min((S[k].x-S[i].x)^2+(S[k].y-S[i].y)^2, dminsq)
20.       k = k + 1
21.   return sqrt(dminsq)
```

a. Complete line 12. [1]

b. Lines 16 to 20 appear to be $\Theta(n^2)$ in the worst case. Explain why this isn't true, and what the worst case efficiency actually is for these lines of code. [3]

c. Superficially, line 7 also appears to be $\Theta(n^2)$. Use pseudocode to show how this can be done with a $\Theta(n)$ algorithm. [4]

Question 12. ToA [8]

Consider the following code:

```
1.   X(A[l .. r]):
2.   p = A[l]
3.   i = l
4.   j = r + 1
5.   repeat:
6.   repeat ++i until A[ i ]>= p
7.   repeat --j until A[ j ]<= p
8.   swap(A[ i ], A[ j ])
9.   until i >= j
10.  swap(A[ i ], A[ j ])
11.  …
12.  return j
```

a. Explain what the algorithm does and give a high-level description of how it accomplishes this. [3]

b. The algorithm contains a bug. Explain what the bug is and how to fix it. [2]

c. Complete the algorithm by filling in line 11. [1]

d. Explain why line 10 is necessary. [2]